Pete Kuzma

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EXPERIENCE

JSPS Postdoctoral Research Fellow
National Astronomical Observatory of Japan — *Oct 2023 - Present, Tokyo, Japan*

- Applied unsupervised machine learning with SKLearn and A/B testing in Python, all maintained with VS Code, to analyse millions of data points, which improved target identification efficiency in multi-dimensional datasets.
- Developed multivariate regression models (scipy, numpy) with Jupyter to extract meaningful patterns from astronomical datasets, ensuring data integrity and optimising selection criteria.
- Automated SQL queries via Python API integrations, improving data retrieval efficiency by 80% and streamlining large-scale data processing.
- Built and led a 10+ member international research team located across four countries, where I delegated roles and tasks to members of appropriate expertise to complete the project.
- Conducted regular meetings with research groups to address future research paths and assess current research progress and directions.
- Presented my machine learning-based results at an international conference to demonstrate the effectiveness of adapting novel techniques to identify faint signals in large datasets.

Post-Doctoral Research Assistant
Univeristy of Edinburgh — *Apr 2020 - Oct 2023, Edinburgh, UK*

- Designed and implemented four Bayesian inference models to analyse large datasets, integrating Python, SQL, and Unix-based tools such as Vim and Emacs, for statistical evaluation.
- Conducted rigorous artificial A/B testing, hypothesis-testing and bootstrapping to assess model accuracy and optimise data-driven decision-making processes.
- Created a scalable automated Python-based ETL pipeline, integrating Table Access Protocols, SQL querying and data manipulation to improve data accessibility and enable efficient query execution for personal and international research projects.
- Mentored, trained and supervised both undergraduate and postgraduate students to develop their research and technical skills.

CSC Commonwealth Rutherford Fellow
University of Edinburgh — *Mar 2018 - Feb 2020, Edinburgh, UK*

- Developed a scalable data analysis pipeline, integrating multiple Unix-based programs with Python wrappers for automation and efficiency.
- Implemented bootstrap analysis and logistic/linear regression to assess data quality and validate model predictions in Python and R.
- Managed database curation for two international collaborations, ensuring structured, high-quality datasets met rigorous research criteria within tight deadlines.

SUMMARY

A self-driven Data Scientist with 10+ years of experience in quantitative analysis, leveraging Python, Bayesian inference, and machine learning to extract insights from large datasets. Skilled in SQL querying, A/B testing, and building scalable data pipelines, I have developed skills in data science, data engineering, and statistical modelling, and I am ready to take on a position more focused exclusively on data science and the challenges therein, where I can leverage my expertise and grow further into a more rounded and experienced data scientist.

LINKS

- Personal Webpage
- GitHub - PeteKuzma
- LinkedIn - Pete Kuzma

EDUCATION

Australian National University
PhD - Astronomy and Astrophysics
Dec 2017

SKILLS

Back End	Python, SQL, R
Front End	HTML
OS	macOS, UNIX/Linux
DevOps	GitHub, GitLab, VS Code
Misc.	TopCat, LaTeX, Microsoft Suite

DATA

Types:
fits ♦ imaging ♦ Big Data ♦ SQL/databases ♦ dat ♦ csv ♦ votable
Abilities:
Mining ♦ Analytics ♦ Engineering ♦ Statistical/numerical modelling ♦ Machine Learning ♦ Management ♦ Quality assessment ♦ ETL pipeline ♦ A/B testing ♦ Visualisation ♦ Prediction ♦ Interpretation ♦ Insight ♦ bootstrapping

Portfolio

Selected Publications

A Pristine View of Galactic Globular Clusters and their Peripheries: Omega Centauri
Kuzma, P.; Ishigaki, M.,
Monthly Notices of the Royal Astronomical Society, 2025, 537, 2752-2762
- Data Manipulation and extraction, ETL pipeline construction.
- Machine learning and k-nearest neighbours in Python.
- Nested sampling.
- Multivariate Gaussian and Akaike Information Criterion fitting.
- Data Visualisation

Kinematics of metallicity populations in Omega Centauri using Gaia Focused Product Release and Hubble Space Telescope
Vernekar, N. ; Lucatello, S; **Kuzma, P.**; Spina, L
Accepted for publication in *Astronomy & Astrophysics*, 2025,
- International collaboration and mentoring.
- Designed a Jupyter notebook for step-by-step spatial transformations of data.

Stellar Cluster in 4MOST
Lucatello, S.; Bragaglia, A.; Vallenari, A.; et al. with **Kuzma, P.**
The Messenger, 2023, 90, 13-16
- Database construction
- Data mining and extraction

Forward and Back: kinematics of the Palomar 5 tidal tails
Kuzma, P.; Ferguson, A. M. N.; Varri, A. L.; Irwin, M. J.; Bernard, E. J.; Tolstoy, E.; Peñarrubia, J.; Zucker, D. B.
Monthly Notices of the Royal Astronomical Society, 2021 512, 315-327
- Likelihood and probability calculation
- Linear and multivariate regression
- Data Visualisation

Kinematics of metallicity populations in Omega Centauri using Gaia Focused Product Release and Hubble Space Telescope
Kuzma, P.; Ferguson, A.; Varri, A.
Monthly Notices of the Royal Astronomical Society, 2021 507, 1127-1137
- ETL pipeline construction, including TAP+, SQL and data manipulation.
- Bayesian Inference and statistical modelling.
- Data Visualisation.

Selected Talks

Extended Structure of Globular Clusters
National Astronomical Observatory of Japan Colloquium, 2024

Extended Structure in Globular Clusters with Gaia
European Astronomical Society Annual Scientific Meeting, 2023

Outer Regions of Globular Clusters with WEAVE
Week of Weave, 2019

The Great Galactic Graveyard of the Milky Way
Dundee Astronomical Society Monthly Meeting, 2022.

Indigenous Australian Astronomy
Astronomical Society of Edinburgh, 2023

Selected Posters

A Pristine Look at Extended Globular Cluster Structure
IAU Symposium 395: Stellar populations in the Milky Way and beyond, 2025.

The Outer Halos of Globular Clusters: NGC 7089 (M2)
Astronomical Society of Australia Annual Scientific Meeting, 2015.

Palomar 5 and its Tidal Tails: A Search for New Members in the Tidal Stream
Astronomical Society of Australia Annual Scientific Meeting, 2014

Personal

When I'm not at a computer, I am an avid player of games, both video and tabletop, and I enjoy pub trivia nights with friends. A lover of football, Eurovision and cats, I like to walk aimlessly to explore new areas and environments, particularly while travelling.

Side/personal projects

- *Restore-Glamaholic*
  Python-based website querying code through API interaction.
- *EPL-Simulation*
  Simulation of a football season.